Lecture 01

Introduction and Linear Systems for Data Engineering

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Lecture Focus

Representing data with vectors and matrices, expressing linear systems as $Ax = b$, identifying solution types, and analyzing systems through determinant, rank, row reduction, and null space. **[Data Eng. Link: Data Storage, Feature Engineering]**

1 Lecture Overview

Linear algebra is the mathematical language used to represent data and perform computations in machine learning and data engineering. Datasets, images, model parameters, and neural-network operations can all be expressed using vectors and matrices.

This lecture introduces basic mathematical objects, systems of linear equations, matrix form, solution types, determinants, rank, row reduction, and null space. Each idea is connected with its computational use in data engineering.

Learning Outcomes

After completing this lecture, students should be able to:

- Explain why linear algebra is important in data engineering and ML.
- Distinguish among scalars, vectors, matrices, and tensors.
- Represent a system of equations in the form $Ax = b$.
- Identify systems having a unique solution, infinitely many solutions, or no solution.
- Explain singular and non-singular matrices.
- Calculate a determinant and matrix rank using Python.
- Perform a basic neural-network matrix operation using NumPy.

1.1 Why Linear Algebra Matters for IT Students

Data engineering and intelligent systems use linear algebra to organize and process large amounts of data efficiently. As an IT professional, you will encounter these concepts in:

- Database query optimization
- Recommendation systems (Netflix, Amazon)

- Search engines (Google PageRank)
- Image and signal processing
- Machine learning and AI applications

IT/System Component	Linear-Algebra Representation
A single data record (e.g., user)	Vector
Complete dataset (e.g., user table)	Matrix
Grayscale image	Matrix of pixel values
Colour image or image batch	Multidimensional tensor
Model parameters (e.g., weights)	Weight vectors or weight matrices
Database join key	Linear transformation
Neural-network layer	Matrix transformation followed by activation
Recommendations (e.g., Netflix)	Matrix factorization (SVD)

Table 1: IT components and their LA representations.

Data Engineering Connection

If a dataset contains m observations and n features, it is commonly stored as a matrix $X \in \mathbb{R}^{m \times n}$. Rows represent observations and columns represent features. Understanding this shape convention is essential when writing data-processing code in SQL, Pandas, or NumPy.

2 Basic Mathematical Objects

2.1 Scalar

A scalar is a single numerical value, for example $a = 5$. In IT, a learning rate, loss value, model accuracy, or database count is a scalar.

2.2 Vector

A vector is an ordered collection of numbers. A student's record (ID, GPA, Age, Credits) may be written as $\mathbf{v} = [101, 3.7, 22, 30]^T$. The vector contains four features, so its dimension is 4.

2.3 Matrix

A matrix is a rectangular arrangement of numbers. Data from multiple students, each with 4 features, can form a data matrix $X \in \mathbb{R}^{3\times 4}$.

2.4 Tensor

A tensor is a multidimensional collection of numbers. A colour image can have shape $h \times w \times 3$, where h and w are its height and width and the last dimension contains red, green, and blue channels.

Notation Check

Symbol	Object	Example Use
a	Scalar	Learning rate, database count
$\mathbf{x}$	Vector	One observation or feature vector
A or X	Matrix	Dataset or weight matrix
T	Tensor	Batch of images, 3D medical data

2.5 Shape Compatibility

For matrix multiplication Ax, the number of columns of A must equal the number of entries in x. If $A \in \mathbb{R}^{m\times n}$ and $x \in \mathbb{R}^n$, then $Ax \in \mathbb{R}^m$. **[Data Eng. Link: Data Wrangling, Data Pipelines]**

[3D Anim: 3D Matrix Multiplication Visualization: Watch dimensions align in real-time.]

3 Systems of Linear Equations

Consider the system:

$$x + y = 10,$$
$$x + 2y = 12.$$

The objective is to find values of x and y that satisfy both equations simultaneously.

3.1 Matrix Representation

The system can be written compactly as:

$$A\mathbf{x} = \mathbf{b} \implies \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 12 \end{bmatrix}.$$

- **Coefficient matrix (A):** Represents features or a linear transformation.
- **Unknown vector (x):** Represents parameters or weights.
- **Target vector (b):** The output or target data.

Solving the system (e.g., by substitution) gives $y = 2$ and $x = 8$.

3.2 Geometric Interpretation

In two variables, each equation represents a line. The solution of the system is the point shared by both lines.

- Lines intersect once $\Rightarrow$ one unique solution.
- Lines coincide $\Rightarrow$ infinitely many solutions.
- Lines are parallel and distinct $\Rightarrow$ no solution.

[3D Anim: 3D Plane Intersection: Visualize unique, infinite, and no-solution cases.]

4 Types of Solutions

Let $[A \mid b]$ denote the augmented matrix and let n be the number of unknowns.

Solution Type	Rank Condition	Meaning
Unique solution	$\text{rank}(A) = \text{rank}([A \mid b]) = n$	Exactly one vector satisfies the system.
Infinitely many	$\text{rank}(A) = \text{rank}([A \mid b]) < n$	At least one variable is free; some information is redund
No solution	$\text{rank}(A) < \text{rank}([A \mid b])$	The equations are inconsistent.

4.1 Singular and Non-Singular Matrices

For a square matrix A:

- **Non-Singular Matrix:** If $\det(A) \neq 0$, then A is non-singular. It has full rank, its inverse exists, and $Ax = b$ has a unique solution.
- **Singular Matrix:** If $\det(A) = 0$, then A is singular. Its inverse does not exist. Depending on $\mathbf{b}$, the system may have infinitely many solutions or no solution.

Common misconception: a zero determinant does not automatically mean "no solution."

5 Row Reduction and Gaussian Elimination

Row reduction transforms a system into a simpler but equivalent system. The permitted elementary row operations are:

1. Interchange two rows.
2. Multiply a row by a non-zero scalar.
3. Add a multiple of one row to another row.

For the system in Section 3, begin with the augmented matrix:

$$\begin{bmatrix} 1 & 1 & 10 \\ 1 & 2 & 12 \end{bmatrix}.$$

Apply $R_2 \leftarrow R_2 - R_1$:

$$\begin{bmatrix} 1 & 1 & 10 \\ 0 & 1 & 2 \end{bmatrix}.$$

The second row gives $y = 2$. Substituting it into the first row gives $x = 8$.

5.1 Echelon Forms

- **Row Echelon Form (REF):** Has all zero rows at the bottom and each leading entry appears to the right of the leading entry above it.
- **Reduced Row Echelon Form (RREF):** Additionally requires every pivot to be 1 and the only non-zero entry in its column.

The RREF of the augmented matrix is:

$$\begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 2 \end{bmatrix}.$$

Data Engineering Connection

Gaussian elimination is a fundamental computational procedure, but software libraries use optimized numerical algorithms. In practice, use a solver such as `np.linalg.solve` instead of explicitly computing $A^{-1}b$.

6 Rank and Null Space

6.1 Rank

The rank of a matrix is the number of linearly independent rows or columns. For $A \in \mathbb{R}^{m\times n}$, $\text{rank}(A) \leq \min(m, n)$. A square matrix has full rank when $\text{rank}(A) = n$.

6.2 Null Space

The null space of A is the set of all vectors satisfying $\mathcal{N}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$. If a non-zero vector belongs to it, then A is singular.

[3D Anim: 3D Null Space Visualization: See the directions that are mapped to zero.]

6.3 Rank-Nullity Preview

For a matrix having n columns,

$$\text{rank}(A) + \text{nullity}(A) = n.$$

7 Python and NumPy Practice

7.1 Representing Vectors and Matrices

```
import numpy as np
x = np.array([2, 4, 6])
A = np.array([ [1, 2], [3, 4] ])
print("Vector:", x)
print("Matrix:\n", A)
print("Matrix shape:", A.shape)
```

7.2 Solving a Linear System

```
A = np.array([ [1, 1], [1, 2] ], dtype=float)
b = np.array([10, 12], dtype=float)
solution = np.linalg.solve(A, b)
print("Solution:", solution)
print("Verification:", A @ solution)
```

Expected output: ‘Solution: [8. 2.]‘ and ‘Verification: [10. 12.]‘

7.3 Determinant and Rank

```
determinant = np.linalg.det(A)
rank = np.linalg.matrix_rank(A)
print("Determinant:", determinant)
print("Rank:", rank)
```

8 Diagnosing Systems with Python

The following function compares the rank of the coefficient matrix with the rank of the augmented matrix.

```
def classify_system(A, b):
    rank_A = np.linalg.matrix_rank(A)
    augmented = np.column_stack((A, b))
    rank_augmented = np.linalg.matrix_rank(augmented)
    variables = A.shape[1]
    if rank_A < rank_augmented:
        return "No solution"
    elif rank_A < variables:
        return "Infinitely many solutions"
    else:
        return "Unique solution"

print(classify_system(A, b))
```

8.1 Neural-Network Connection

A basic neural-network layer calculates $\mathbf{z} = W\mathbf{x} + \mathbf{b}$, where $\mathbf{x}$ is the input vector, W is the weight matrix, $\mathbf{b}$ is the bias vector, and $\mathbf{z}$ is the output before activation.

```
import numpy as np
x = np.array([0.8, 0.3, 0.5])
W = np.array([
    [0.4, 0.7, 0.2],
    [0.6, 0.1, 0.9]
])
bias = np.array([0.1, 0.2])
z = W @ x + bias
relu_output = np.maximum(0, z)
print("Linear output:", z)
print("After ReLU:", relu_output)
```

[ML/DS Connection: Neural network layer as an affine transformation.]

9 Review and Independent Practice

Practice Tasks

1. Represent the system $2x + y = 7$ and $x - y = 2$ in the form $Ax = b$.
2. Solve the system manually using row reduction, then verify with `np.linalg.solve`.

3. Find the determinant and rank of $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. Is it singular?

4. For the same matrix, compare $\mathbf{b}_1 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$. Classify both systems.

5. Change the values in the neural-layer weight matrix W. Record how the output changes.

6. Create a 4×3 NumPy matrix. State how many observations and features it could represent.

Key Takeaways

- Data and parameters are commonly represented using vectors, matrices, and tensors.
- A linear system is written compactly as $Ax = b$.
- Rank measures independent information and helps determine the number of solutions.
- A non-singular square matrix has a unique solution.
- The null space describes input directions mapped to zero.
- NumPy provides efficient tools for solving and analysing linear systems.
- A neural-network layer is built from the matrix transformation $Wx + b$.

Key Terms

Scalar, Vector, Matrix, Tensor, Linear system, Augmented matrix, Determinant, Rank, Null space, Singular matrix, Gaussian elimination, Neural layer.